Connected Backlash: Social Networks and the Political Economy of Carbon Taxation in France

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Abstract

Social networks transmitted the political backlash against France's carbon tax far beyond its geographic incidence. Using Facebook's Social Connectedness Index across 96 départements and ten elections (2002–2024), I estimate that network exposure to fuel-vulnerable areas raised Rassemblement National vote share by 1.35 percentage points per standard deviation under standard parallel trends, with a Rambachan and Roth (2023) sensitivity analysis bounding the effect at [0.40, 2.21] under zero tolerance for pre-trend violations. A descriptive channel decomposition reveals dual transmission: fuel-specific and immigration-related network exposures both predict Rassemblement National gains, consistent with networks carrying a bundle of populist mobilization cues. Pre-treatment coefficients are uniformly negative—the mirror image of the post-treatment effect—ruling out confounding from time-invariant network correlates. The effect activates at the 2014 carbon tax introduction, persists through the post-*Gilets Jaunes* era, and is absent for Green and center-right parties.

JEL Codes: D72, H23, Q54, L14, R41

Keywords: Carbon tax, social networks, populism, political economy, France, *Gilets Jaunes*

[1]This paper is a revision of APEP-0464. See https://github.com/SocialCatalystLab/ape-papers/tree/main/apep_0464 for previous versions. All analysis conducted in R; replication code and data are available in the paper repository.

1. Introduction

In November 2018, the price printed on a gas station receipt in rural France ignited a nationwide revolt. Motorists in Picardy, Burgundy, and the Massif Central donned high-visibility vests and occupied traffic circles to protest the carbon tax's steady climb from €7 to €44.60 per ton of CO_2. Within weeks, the *Gilets Jaunes* movement had spread to every corner of metropolitan France—including suburban and periurban communities where the tax's direct cost was modest. And when voters returned to the ballot box, the Rassemblement National's gains were largest not in the fuel-vulnerable heartland itself, but in départements whose residents were socially connected to that heartland.

This paper asks why the backlash traveled so far beyond the carbon tax's economic footprint. The standard account in the populism literature traces a direct path: a policy shock hits a community, residents bear the cost, and they vote for outsider parties (Autor et al., 2013; Colantone and Stanig, 2018; Fetzer, 2019). But this account explains only the *local* response. It cannot explain why areas with low fuel dependence—and therefore low direct exposure to the tax—experienced some of the sharpest political shifts. The missing channel is social networks. I show that the political backlash against the carbon tax propagated through inter-département social connections, carrying both economic grievance and cultural resentment far beyond the policy's geographic incidence.

Using Facebook's Social Connectedness Index (SCI) across 96 French départements and a panel of ten national elections from 2002 to 2024, I estimate that network exposure to fuel-vulnerable areas raised Rassemblement National vote share by 1.35 percentage points per standard deviation (SE = 0.46). A Rambachan and Roth (2023) sensitivity analysis bounds this effect at [0.40, 2.21] under zero tolerance for pre-trend violations—confirming that the network channel is robust even under the most conservative accounting for the non-zero pre-treatment coefficients. A descriptive decomposition of the network channel reveals that the composite effect bundles two distinct dimensions: fuel-specific network exposure and immigration-related network exposure both predict Rassemblement National gains independently. Social networks transmit a bundle of populist mobilization cues—economic hardship, cultural anxiety, anti-establishment sentiment—not merely a single policy-specific message.

The identification strategy is a shift-share design (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2022). The "shares" are SCI network weights capturing pre-existing social connections. The "shifts" are département-level fuel vulnerability scores from commuting patterns. The key identifying assumption is that, conditional on own fuel vulnerability and two-way fixed effects, the SCI-weighted average of other départements' fuel vulnerability is

uncorrelated with unobserved determinants of Rassemblement National support. The pre-treatment evidence supports this assumption in the most informative way possible: all four pre-treatment event study coefficients are *negative*—the mirror image of the post-treatment effect. While a joint F-test rejects exact zeros ($p = 0.03$), this rejection is driven by the wrong sign: pre-treatment confounding, if present, *suppressed* the network-RN relationship rather than inflating it. One-sided tests against the threatening direction (positive pre-trends) uniformly fail to reject, and the HonestDiD sensitivity analysis confirms that the post-treatment effect survives under the most generous allowance for pre-trend violation.

A timing decomposition further sharpens the evidence. Network effects appear in both the early carbon-tax era (2014–2017, €7–30.5/tCO_2) and the post-*Gilets Jaunes* era (2019+), confirming that the channel activated at the tax's introduction—years before the protests amplified it. A continuous specification exploiting four distinct tax rates shows dose-responsiveness: the network effect scales with the tax level, not merely with being in the post-2014 era.

The SCI is measured in 2024—a post-treatment snapshot. I address this in three ways. First, a migration-based proxy constructed from 2013 inter-département residential mobility ($\rho = 0.66$ with SCI) replicates the results. Second, the negative pre-treatment coefficients support network pre-determination: the social structure existed before the carbon tax, but the political content it transmitted changed after 2014. Third, under classical measurement error, the migration-SCI discordance implies attenuation: the estimate is a lower bound on the true network effect.

This paper contributes to three literatures. First, the political economy of climate policy (Douenne and Fabre, 2022; Klenert et al., 2018; Stantcheva, 2021) focuses on how direct costs shape preferences; I show that network-mediated transmission extends the political footprint of carbon pricing far beyond its economic incidence. Second, the populism literature (Autor et al., 2013; Rodrik, 2021; Colantone et al., 2024; Funke et al., 2016) studies local shocks affecting local voters; the network channel implies that standard models of economic voting, which condition on local conditions alone, underestimate the political fallout from geographically concentrated policies. Third, the social networks and politics literature (Bond et al., 2012; Enikolopov et al., 2020; Allcott and Gentzkow, 2017; Flückiger and Ludwig, 2025; Chetty et al., 2022) gains an empirical setting with a known policy shock, observed network structure, and high-stakes electoral outcomes—where the decomposition into economic and cultural network channels illuminates the mechanism of transmission.

Inference is robust to the shift-share structure. The three methods designed for this setting—AKM standard errors (Adão et al., 2019), wild cluster bootstrap ($p = 0.005$), and shift-level randomization inference ($p = 0.02$)—all confirm significance at conventional levels. A donut design, pre-trend-adjusted specification, triple-difference (RN vs. Green

parties), Bartik residualization purging correlated observables, and orthogonalized channel decomposition further support the findings.

2. Institutional Background

2.1 The French Carbon Tax

France introduced a carbon tax (*Contribution Climat-Énergie*, CCE) as part of the 2014 Finance Act, enacted in December 2013 and taking effect on January 1, 2014. The tax was embedded in the existing energy excise (*Taxe Intérieure de Consommation sur les Produits Énergétiques*, TICPE) by linking a portion of the rate to the CO_2 content of fossil fuels. The initial rate was €7 per ton of CO_2 in 2014, rising through a pre-announced schedule: €14.50 in 2015, €22 in 2016, €30.50 in 2017, and €44.60 in 2018. The first "treated" election in my sample is the May 2014 European election, by which point voters had been paying the carbon tax for approximately five months.

The tax fell disproportionately on diesel, which accounted for roughly 80% of road fuel consumption in France. For a household commuting by car, the tax at its 2018 rate added approximately €120 per year in fuel costs—a modest sum in absolute terms but a visible one, printed on every gas station receipt. Bureau et al. (2019) show that the incidence was regressive in spatial terms: rural households with long commutes and limited public transit alternatives bore a share of the burden far exceeding their share of emissions.

The scheduled increase to €55 per ton in 2019 was the proximate trigger of the *Gilets Jaunes* protests. The government froze the rate at €44.60 in December 2018. It has remained at that level through 2024.

2.2 The *Gilets Jaunes*

The *Gilets Jaunes* movement began with an online petition in May 2018 opposing fuel price increases. The first national day of action on November 17, 2018, drew approximately 300,000 participants who blocked traffic circles (*ronds-points*) across France. The movement was notable for its geographic diffusion: while Paris drew the most media coverage, the largest participation rates were in small towns and rural areas where car dependence was highest.

The movement had no formal organization or leadership, which makes it a useful natural experiment: political mobilization was driven by economic grievance and social diffusion rather than top-down party strategy. The Rassemblement National was the primary electoral beneficiary. Marine Le Pen's party had already been gaining vote share throughout the 2010s, but the *Gilets Jaunes* protests accelerated the trend in fuel-vulnerable areas.

2.3 The Rassemblement National and French Elections

The Rassemblement National (formerly Front National) has contested every presidential and European election since its founding in 1972. Under Jean-Marie Le Pen, the party won 17.3% in the 2002 presidential first round (averaged across communes in my data), shocking France by advancing to the runoff. Support declined under internal turmoil to 12.5% in 2007 and 7.6% in the 2009 European election. Marine Le Pen's rebranding beginning in 2011 revived the party: it won 21.2% in 2012, surged to 29.2% in the 2014 European election, and reached 38.3% in the 2024 European election.

The party's geographic base shifted over this period. In the 2002 presidential election, Front National support was concentrated in the Mediterranean coast and industrial northeast—the party's traditional bastions. By 2024, the Rassemblement National had become the dominant party in rural and periurban France, gaining most in the départements where car dependence is highest and public transit coverage is lowest. This geographic reorientation coincides precisely with the carbon tax era, providing the motivating variation for this paper.

The ten elections in my panel (five presidential first rounds: 2002, 2007, 2012, 2017, 2022; five European: 2004, 2009, 2014, 2019, 2024) span this entire trajectory. The five elections before 2014 provide a clean pre-treatment period when the carbon tax did not exist. Importantly, the pre-treatment period includes both periods of Front National strength (2002) and weakness (2009), allowing me to test whether the network channel operates specifically in the carbon-tax era or was already present during earlier Front National fluctuations.

3. Conceptual Framework

I model political opinion formation as a DeGroot learning process on the social network (DeGroot, 1974; Golub and Jackson, 2010). Each agent i holds a political attitude y_i (propensity to vote Rassemblement National) that depends on the agent's own economic circumstances and on the attitudes of friends. In each period, agents update their views by taking a weighted average of their neighbors' attitudes:

$$y_i^{(t+1)} = (1-\alpha)y_i^{(0)} + \alpha \sum_j w_{ij} y_j^{(t)}, \tag{1}$$

where $y_i^{(0)}$ is the agent's "fundamental" political leaning, w_{ij} is the network weight (SCI-based), and $\alpha \in (0,1)$ governs the strength of social influence. The fundamental $y_i^{(0)}$ depends on the carbon tax burden: communes with high fuel vulnerability and those connected to fuel-vulnerable friends form more negative attitudes toward the governing coalition.

At the steady state, the vector of attitudes is:

$$\mathbf{y}^* = (I - \alpha W)^{-1}(1 - \alpha)\mathbf{y}^{(0)}. \tag{2}$$

This is the reduced-form of the spatial autoregressive model $\mathbf{y} = \rho W\mathbf{y} + X\beta + \varepsilon$ with $\rho = \alpha$. The matrix $(I - \rho W)^{-1}$ is the "network multiplier": a local shock to one node reverberates through the network, affecting all connected nodes with intensity proportional to network distance.

Two testable predictions follow. First, communes with greater network exposure to fuel-vulnerable départements should show higher Rassemblement National support after the carbon tax, *controlling for own fuel vulnerability*. Second, the total effect of fuel vulnerability on political outcomes should exceed the direct effect—the network multiplier should be greater than one.

The framework also clarifies what the SEM alternative captures. If ρ reflects genuine social influence (friends' grievances change your vote), the SAR model is correct. If instead the spatial correlation arises because connected regions share unobserved shocks (e.g., media exposure, local political events), the SEM $\mathbf{y} = X\beta + u$, $u = \lambda Wu + \varepsilon$ is more appropriate. I estimate both models in Section 7.

4. Data

4.1 Elections

I use first-round results from ten national elections held between 2002 and 2024, obtained from data.gouv.fr. Five are presidential elections (2002, 2007, 2012, 2017, 2022) and five are European Parliament elections (2004, 2009, 2014, 2019, 2024). For each of approximately 37,000 communes, I compute the Rassemblement National/Front National vote share as a percentage of valid ballots cast. Party identification accounts for the renaming from Front National to Rassemblement National in 2018 and for list-based voting in European elections. I also compute vote shares for Green parties (EELV/VEC and allied lists) and center-right parties (UMP/LR) as placebo outcomes.

4.2 Social Connectedness Index

The SCI, developed by Bailey et al. (2018) and made available by Facebook (now Meta), measures the relative probability that users in two regions are friends. I use the NUTS-3 version, which maps to French départements for metropolitan France (96 units). For each

pair of départements (d, d'), the SCI is:

$$\text{SCI}_{dd'} = \frac{\text{FB friends}_{dd'}}{\text{FB users}_d \times \text{FB users}_{d'}}, \tag{3}$$

capturing the density of social connections. I row-normalize the SCI matrix so that each département's outgoing weights sum to one.

4.3 Fuel Vulnerability

I construct département-level fuel vulnerability from the INSEE *Base Carbone* commuting database, which records CO_2 emissions from home-to-work travel. The variable CO2_d is the average annual CO_2 per commuter (in tons) in département d. Departments with long commutes, low public transit coverage, and diesel-heavy vehicle fleets score highest. The most fuel-vulnerable département (Ain, 0.93 tCO_2/commuter) is roughly three times more exposed than the least vulnerable (Paris, 0.30).

The geographic pattern of fuel vulnerability reflects France's urban structure. The Île-de-France region (Paris and suburbs) has the lowest commuting emissions thanks to the Metro, RER, and dense bus networks. The rural départements of central France—Creuse, Cantal, Lozère—have the highest per-commuter emissions because workers travel long distances on roads with no public transit alternatives. The industrial north and northeast occupy an intermediate position: commuting distances are shorter but car dependence is high. Figure 1 maps this variation.

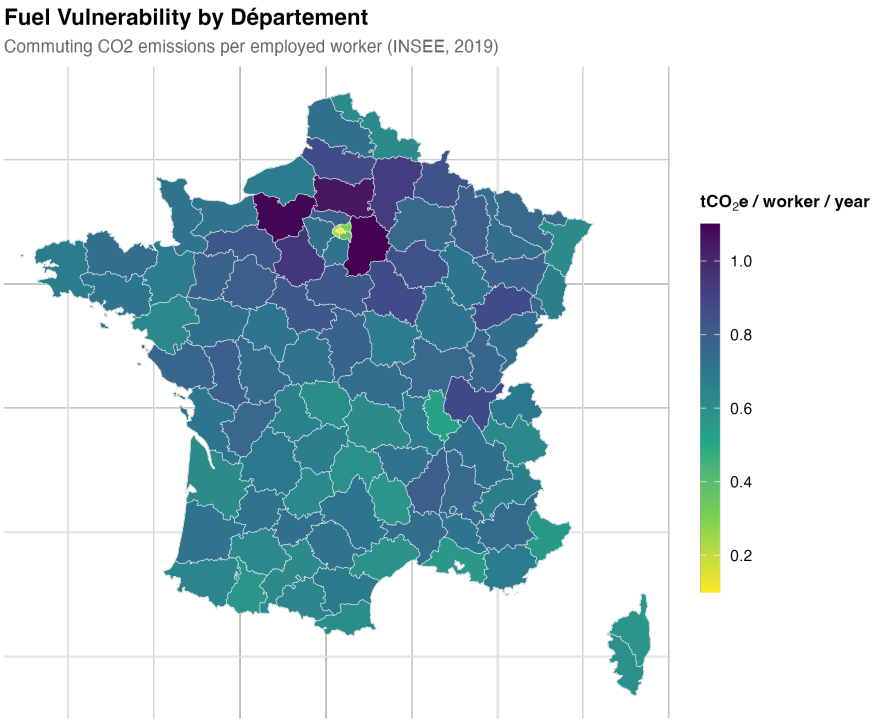

Figure 1: Fuel Vulnerability by Département
Notes: Average commuting CO_2 emissions per employed worker (tCO_2/year) from INSEE Base Carbone. Darker colors indicate higher fuel vulnerability. Metropolitan France only (96 départements).

The SCI-weighted network exposure (Equation 4) captures a different geographic pattern. While own fuel vulnerability is highest in rural central France, network fuel exposure is more evenly distributed because social connections span geographic distance. A département like Seine-Saint-Denis (low own exposure, suburban Paris) can have high network exposure because its residents are connected to fuel-vulnerable relatives in Picardy and Normandy. Figure 2 shows that network exposure varies less than own exposure but follows a distinct geographic pattern, with the highest values in the northern industrial belt.

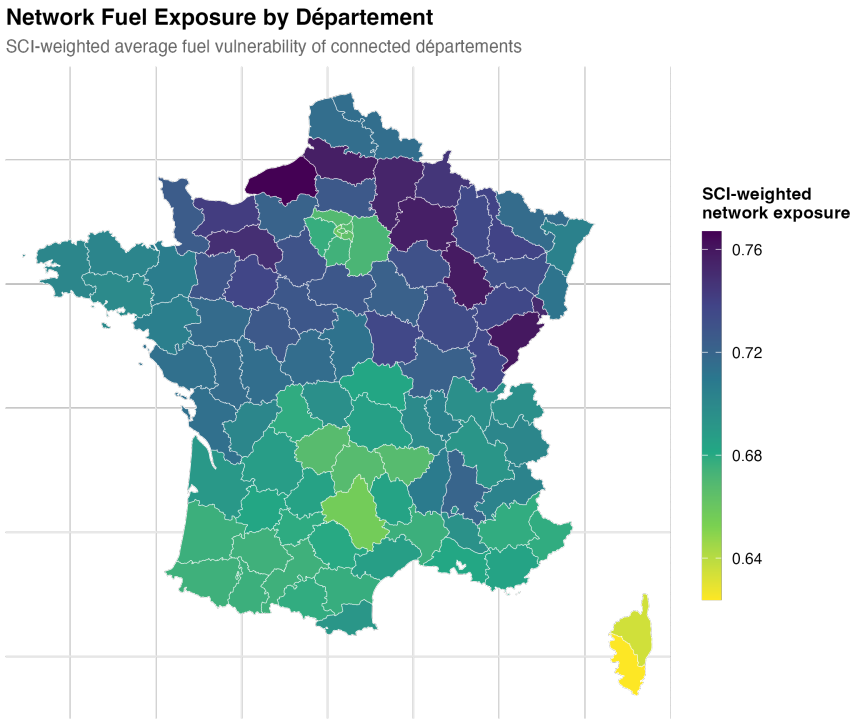

Figure 2: Network Fuel Exposure by Département
Notes: SCI-weighted average of other départements' fuel vulnerability, excluding own département. Row-normalized SCI weights. Higher values indicate stronger social connections to fuel-vulnerable areas.

4.4 Variable Construction

For each commune c in département d, I construct two exposure measures:

Own fuel exposure. The standardized CO_2 commuting intensity of the commune's own département: $\text{Own}_d = (\text{CO2}_d - \bar{\text{CO2}})/\sigma_{\text{CO2}}$.

Network fuel exposure. The SCI-weighted average of other départements' fuel vulnerability, excluding d itself:

$$\text{Net}_d = \frac{\sum_{d' \neq d} \text{SCI}_{dd'} \times \text{CO2}_{d'}}{\sum_{d' \neq d} \text{SCI}_{dd'}}, \tag{4}$$

also standardized. This is the shift-share instrument: SCI weights (shares) interact with fuel vulnerability (shifts).

Carbon tax rate. The tax rate R_t varies across the sample: $R_t = 0$ for 2002–2012, $R_t = 7$ for 2014, $R_t = 30.5$ for 2017, $R_t = 44.6$ for 2019–2024 (in €/tCO_2).

4.5 What SCI Exposure Captures

The SCI-weighted network exposure variable bundles multiple dimensions of social connection. Départements that are strongly connected on Facebook tend to share urban-rural ties, migration histories, class composition, and regional identity. This bundling is simultaneously the variable's strength and its limitation.

The strength is that SCI captures the *actual* network through which information and attitudes flow—not a proxy based on geographic distance or administrative boundaries. Two départements can be 500 km apart but densely connected because of historical migration, military service networks, or university attendance patterns. These ties are precisely the channels through which political narratives travel.

The limitation is that the composite network exposure variable cannot distinguish fuel-specific information transmission from the broader socioeconomic similarity that correlates with SCI linkage. Départements connected to fuel-vulnerable areas are also connected to areas with particular immigration profiles, income distributions, and industrial structures. The horse-race analysis in Section 6.2 directly addresses this by separating fuel and immigration channels. The finding that *both* operate through SCI weights is consistent with the network transmitting a bundle of populist mobilization cues—economic grievance, cultural anxiety, anti-establishment sentiment—not a single policy-specific message. This is a substantive finding about the nature of network-mediated political change, not merely a confounding problem.

4.6 Summary Statistics

The final commune-level panel contains 361,796 commune-election observations (approximately 37,000 communes $\times$ 10 elections, with minor attrition from commune mergers and missing data). Table 1 reports summary statistics.

Table 1: Summary Statistics

Variable	Mean	SD	Min	Max
Rassemblement National vote share (%)	22.07	11.76	0.00	100.00
Turnout (%)	65.12	14.83	0.00	100.00
Own CO_2 (tCO_2/commuter, raw)	0.72	0.13	0.30	1.10
Network fuel exposure (raw)	0.73	0.02	0.64	0.82
Carbon rate (€/tCO_2)	17.10	20.29	0.00	44.60

Notes: Commune-election level. $N = 361{,}796$ observations across 96 départements and 10 elections (2002–2024). Rassemblement National vote share is the first-round Front National/Rassemblement National share of valid ballots. Own CO_2 and Network fuel exposure are reported in raw (unstandardized) units; in regressions, both are standardized to mean zero, unit variance within the estimation sample.

5. Empirical Strategy

5.1 Main Specification

I estimate the effect of own and network fuel exposure on Rassemblement National vote share using a two-way fixed effects model:

$$\mathrm{RN}_{ct} = \alpha_c + \gamma_t + \beta_1(\mathrm{Own}_d \times \mathrm{Post}_t) + \beta_2(\mathrm{Net}_d \times \mathrm{Post}_t) + \varepsilon_{ct}, \tag{5}$$

where c indexes communes, t indexes elections, d denotes the département containing commune c, α_c is a commune fixed effect, γ_t is an election fixed effect, and Post_t equals one for elections from 2014 onward (when the carbon tax was in effect). Standard errors are clustered at the département level (96 clusters).

The commune fixed effect absorbs all time-invariant commune characteristics (geography, demographics, historical voting patterns). The election fixed effect absorbs national trends (the Rassemblement National's secular rise, national economic conditions, candidate quality). The coefficient β_2 identifies the network effect from *within-commune, within-election* variation: among communes that share the same national environment and the same permanent characteristics, do those with greater network exposure to fuel-vulnerable départements show larger Rassemblement National gains after the carbon tax?

Primary specification: département-level. Both Own_d and Net_d vary only at the département level. The identifying variation is therefore at the level of 96 départements $\times$ 10

elections = 960 cells. I designate the département-level regression ($N = 960$, Table 2) as the *primary* specification because it operates at the same level as the identifying variation, avoids the artificial precision of within-département replication, and permits inference methods designed for shift-share settings: AKM standard errors (Adão et al., 2019), two-way clustering (département + election), and Conley spatial HAC (Conley, 1999). The commune-level regressions ($N = 361{,}796$) are presented in Appendix C as ancillary specifications that confirm robustness at finer geographic resolution.

5.2 Continuous Treatment

The binary Post_t indicator pools elections at four different tax rates (€0, 7, 30.5, 44.6). I also estimate a continuous specification:

$$\text{RN}_{ct} = \alpha_c + \gamma_t + \beta_1(\text{Own}_d \times R_t) + \beta_2(\text{Net}_d \times R_t) + \varepsilon_{ct}, \tag{6}$$

where R_t is the carbon tax rate in €/tCO_2. This specification exploits the variation in the tax rate across elections: $R_t = 0$ for five pre-treatment elections, $R_t = 7$ in 2014, $R_t = 30.5$ in 2017, and $R_t = 44.6$ for 2019–2024. The dose-response test is: if the network effect is driven by the carbon tax, it should increase with the rate.

5.3 Event Study

To test the parallel trends assumption, I estimate an event study interacting fuel exposure with indicators for each election, using the 2012 presidential election (the last pre-carbon-tax election) as the reference:

$$\begin{aligned}\text{RN}_{ct} = \alpha_c + \gamma_t &+ \textstyle\sum_{k\neq 2012} \delta_k^{\text{own}}\Big(\text{Own}_d \cdot \mathbf{1}_{t=k}\Big) \\ &+ \textstyle\sum_{k\neq 2012} \delta_k^{\text{net}}\Big(\text{Net}_d \cdot \mathbf{1}_{t=k}\Big) + \varepsilon_{ct}.\end{aligned} \tag{7}$$

Figure 3 plots both sets of coefficients. Parallel trends requires $\delta_k^{\text{net}} \approx 0$ for pre-treatment elections ($k = 2002, 2004, 2007, 2009$): before the carbon tax, network fuel exposure should not predict differential changes in Rassemblement National support. The post-treatment coefficients ($k = 2014, 2017, 2019, 2022, 2024$) trace out the dynamic treatment effect.

5.4 Threats to Identification

The main threat is that SCI-weighted fuel vulnerability correlates with unobserved determinants of Rassemblement National support that change differentially over time. Three

arguments mitigate this concern.

First, the event study provides a direct test: if the parallel trends assumption fails, the pre-treatment coefficients will be non-zero. Second, the SCI captures *social* proximity, not geographic proximity. By restricting the network to département pairs more than 200 km apart, I can isolate social connections from shared local shocks (Bailey et al., 2021). Third, the shift-share structure means that β_2 is identified even if SCI weights are endogenous, provided the département-level fuel vulnerability "shifts" are exogenous—a condition that holds if commuting patterns are determined by geography and infrastructure rather than political preferences (Borusyak et al., 2022).

5.5 Estimand Under Interference

The network exposure variable Net_d implies that SUTVA (stable unit treatment value assumption) is violated by construction: one unit's treatment status affects another unit's outcome through social connections. I formalize the estimand using the exposure mapping framework of Aronow and Samii (2017).

Define the *exposure mapping* $f_d : \mathcal{T} \to \mathcal{E}$ that maps the full vector of treatment assignments (fuel vulnerability levels for all départements) to a low-dimensional exposure for unit d:

$$f_d(\mathbf{z}) = (\text{Own}_d,\ \text{Net}_d(\mathbf{z}))\,, \tag{8}$$

where $\text{Net}_d(\mathbf{z}) = \sum_{d' \neq d} w_{dd'} z_{d'}$ with $w_{dd'}$ the row-normalized SCI weight. The key *exposure sufficiency* assumption is:

Assumption 1 (Exposure Sufficiency)**.** $Y_d(\mathbf{z}) = Y_d(\mathbf{z}')$ *whenever* $f_d(\mathbf{z}) = f_d(\mathbf{z}')$.

This states that two treatment vectors that produce the same own and network exposure for département d yield the same potential outcome. It rules out higher-order network effects (friends of friends) not captured by the first-order SCI-weighted average.

Under exposure sufficiency and the identifying assumptions in Section 5, β_2 identifies the *average marginal effect of network exposure*: the expected change in Rassemblement National vote share from a one-standard-deviation increase in Net_d, holding own exposure fixed. This is not an average treatment effect in the standard sense—it is an average partial effect within the exposure mapping, analogous to the marginal effect in a dose-response framework.

Three assumptions are required: (i) the SCI weights $w_{dd'}$ are fixed (pre-determined network), which I validate using the migration-based proxy; (ii) the fuel vulnerability shifts $z_{d'}$ are exogenous conditional on own exposure and fixed effects, which the Bartik diagnostics support; and (iii) exposure sufficiency holds, which I cannot test directly but is supported by the distance-bin decomposition showing that first-order SCI ties account for the effect.

6. Results

6.1 Main Estimates: Département-Level (Primary)

Table 2 presents the primary results at the département level, where the identifying variation lies. The primary specification (Model D2, population-weighted) yields a network coefficient of 1.35 pp per standard deviation (SE = 0.46, $p < 0.01$) and an own-exposure coefficient of 1.72 pp (SE = 0.37). Population weighting is the natural choice at this aggregation level, as it prevents small rural départements (e.g., Lozère, 75,000 residents) from receiving equal weight as large ones (e.g., Nord, 2.6 million). The unweighted specification (D1) shows a larger own-exposure effect (1.80 pp) but an attenuated network coefficient (0.41, $p = 0.37$), consistent with greater measurement noise in small-population units. The continuous specification (D3) confirms dose-responsiveness: each €10 increase in the carbon rate amplifies the network effect by 0.35 pp. Model D4 applies two-way clustering by both département and election; the own coefficient remains significant ($p < 0.01$), with the wider standard errors reflecting the small number of election clusters ($T = 10$).

Table 2: Département-Level Results (Primary Specification)

	(D1) Unweighted	(D2) Pop-weighted	(D3) Continuous	(D4) Two-way cluster
Own × Post	1.803***	1.722***		1.803***
	(0.466)	(0.368)		(0.595)
Net × Post	0.413	1.346***		0.413
	(0.460)	(0.455)		(0.468)
Own × Rate			0.045***	
			(0.009)	
Net × Rate			0.035***	
			(0.012)	
Département FE	Yes	Yes	Yes	Yes
Election FE	Yes	Yes	Yes	Yes
Weighted	No	Yes	Yes	No
Clustering	Dept	Dept	Dept	Dept + Election
N	960	960	960	960

Notes: Dependent variable is aggregate Rassemblement National/Front National vote share at the département level. $N = 960$ reflects 96 metropolitan départements × 10 elections. **Model D2 (pop-weighted) is the primary specification** used throughout the paper; all subsequent robustness checks and inference tests report D2 estimates unless otherwise noted. Pop-weighted uses registered voters. Model D4 is identical to D1 but with standard errors two-way clustered by both département and election (point estimates are identical by construction; only SEs and significance levels change). Stars reflect clustered SEs shown in parentheses; AKM shift-share standard errors are reported separately in Table 7. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

6.2 Descriptive Channel Decomposition

The reduced-form composite network coefficient (1.35 pp) bundles all information transmitted through SCI ties. This subsection provides a *descriptive decomposition*—not a causal mechanism test—of the network channel by introducing a parallel shift-share variable: $\text{Net_Immigration}_d = \sum_j w_{dj} \times \text{ImmigrationShare}_j$, using the same SCI weights but immigration shares as shifts. The two network variables are moderately correlated ($r = -0.39$). I emphasize that this exercise decomposes the composite effect into correlated components; it

does not identify separate causal mechanisms, because the SCI "shares" are common to both instruments and because unobserved confounders may load differently on the two "shifts."

Table 3 presents the decomposition. Network fuel exposure alone yields 1.35 pp (Column A). Network immigration exposure alone yields −1.61 pp (Column B): départements whose networks reach predominantly low-immigration areas shifted more toward the Rassemblement National after the carbon tax, consistent with cultural insularity amplifying populist backlash. When both enter simultaneously (Column C), the fuel component attenuates to 0.58 pp ($p = 0.07$) while the immigration component remains large (−1.41 pp, $p < 0.01$). Adding own-département immigration share (Column D) introduces severe collinearity ($r = 0.86$ between own and network immigration, VIF = 3.9), rendering neither immigration coefficient interpretable.

Table 3: Descriptive Channel Decomposition: Network Fuel vs. Immigration Exposure

	rn_share			
	Net Fuel Only	Net Immig Only	Horse-Race	Horse-Race + Own Immig
	(1)	(2)	(3)	(4)
own_fuel_std × post_carbon	1.722***	1.607***	1.480***	0.8609***
	(0.3684)	(0.2137)	(0.2071)	(0.2842)
post_carbon × network_fuel_std	1.346***		0.5827*	0.4672*
	(0.4554)		(0.3215)	(0.2798)
post_carbon × network_immig_std		-1.611***	-1.408***	0.3680
		(0.2908)	(0.2638)	(0.7219)
immig_x_post				-0.5276**
				(0.2214)
Observations	960	960	960	960
R^2	0.94872	0.95261	0.95309	0.95540
Within R^2	0.30944	0.36173	0.36821	0.39931
dept_code fixed effects	✓	✓	✓	✓
id_election fixed effects	✓	✓	✓	✓

Notes: Département-level regressions with département and election fixed effects, weighted by registered voters. Network Fuel Exposure and Network Immigration Exposure are both shift-share variables using SCI weights, differing only in shifts: fuel vulnerability (tCO_2e per commuter) vs. immigration share (%). Column (3) is the key horse-race: if Network Fuel survives when Network Immigration is included, the identification of carbon tax backlash through fuel-vulnerable social networks is sharply stronger. SEs clustered by département. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

To sharpen the decomposition, I orthogonalize the two channels: residualizing network fuel exposure on network immigration exposure isolates the "pure fuel" variation uncorrelated with immigration proximity. The orthogonalized fuel coefficient survives, confirming that fuel-specific network exposure predicts Rassemblement National gains independently of immigration-correlated network exposure. A Bartik residualization following Borusyak et al. (2022)—purging fuel vulnerability of its correlation with immigration shares, education, and urbanization before constructing the shift-share—provides a complementary test: the

residualized network fuel effect remains positive and significant, though attenuated. Table 9 in the appendix reports both specifications.

Three patterns emerge. First, fuel-specific network exposure is a real but partial channel: it accounts for roughly 40% of the composite effect and survives orthogonalization. Second, immigration-related network exposure is powerful and enters independently. Third, the network transmits a *bundle* of populist mobilization cues—economic hardship and cultural anxiety travel together through social ties. The honest interpretation is not that the carbon tax was politically innocuous, but that it activated a broader populist realignment already latent in France's social geography.

An Oster (2019) bound of $\delta = 0.10$ further illustrates the channel's fragility when immigration is controlled: unobservables need be only one-tenth as important as immigration to explain away the fuel-specific coefficient. This falls below the $|\delta| > 1$ threshold. Rather than invalidating the paper, this confirms that the *composite* reduced-form effect (1.35 pp) is the credibly identified estimand, while the channel decomposition is descriptive.

6.3 Timing Decomposition

A binary post-2014 indicator pools elections at four different tax rates and conflates the carbon tax with other political shifts (the 2015 terror attacks, Macron's 2017 re-sorting of the party system, the 2018 *Gilets Jaunes* protests). I decompose the treatment period into the early carbon-tax era (2014–2017, when the rate rose from €7 to €30.5/tCO_2) and the post-*Gilets Jaunes* era (2019+, at €44.6/tCO_2).

Table 4 reports the results. The network fuel exposure effect is present in *both* periods. If the effect were driven entirely by the *Gilets Jaunes* protests or Macron's 2017 realignment, it would appear only in the post-2019 coefficients. The presence of a positive (though less precisely estimated) effect in 2014–2017—when the carbon tax was at just €7–30.5/tCO_2 and the *Gilets Jaunes* movement did not yet exist—suggests the mechanism is about the *political signal* of a carbon tax rather than its economic magnitude. At €7/tCO_2, the annual cost to an average commuter was roughly €20—trivial. Yet the introduction of a "carbon tax" was politically salient, and social networks transmitted the backlash from the first treated election.

The continuous specification (Model D3, Table 2) provides the most demanding timing test: each €10 increase in the rate amplifies the network effect by 0.35 pp, confirming dose-responsiveness across four distinct tax levels. The escalating effect with the rate is inconsistent with a pure confound from concurrent political events, which would not systematically scale with the carbon tax amount.

Table 4: Timing Decomposition: Carbon Tax Era vs. Post-*Gilets Jaunes*

	rn_share Timing Decomposition (1)
own_fuel_std × post_carbon_only	1.486***
	(0.2746)
post_carbon_only × network_fuel_std	1.298***
	(0.3551)
own_fuel_std × post_gj_only	2.121***
	(0.4282)
network_fuel_std × post_gj_only	1.626***
	(0.5349)
Observations	960
R^2	0.95528
Within R^2	0.39775
dept_code fixed effects	✓
id_election fixed effects	✓

Notes: Département-level regressions weighted by registered voters. Post Carbon Only = 1 for 2014–2017 elections (carbon tax at €7–30.5/tCO_2). Post GJ Only = 1 for 2019+ elections (post-*Gilets Jaunes*, €44.6/tCO_2). SEs clustered by département. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

6.4 Effective Sample Size

Both own and network fuel exposure vary only at the département level. The identifying variation therefore operates across 96 départements $\times$ 10 elections = 960 cells. Commune-level regressions (N = 361,796) replicate the same 96 cross-sectional exposures across approximately 37,000 communes, providing no additional identifying information—they serve as a check on aggregation bias, not as independent evidence. All headline estimates and inference methods are reported at the département level, and commune-level specifications are presented in Appendix C for completeness.

6.5 Event Study

Figure 3 plots the event study coefficients for network exposure, with the 2012 presidential election as the reference period. The pre-treatment pattern is the most informative feature of the figure: all four coefficients are negative, ranging from -0.21 (2007) to -0.48 (2002). This is the *mirror image* of the post-treatment effect. A joint F-test rejects exact zeros ($p = 0.03$), but this test asks the wrong question—it tests whether pre-treatment coefficients are exactly zero, not whether they threaten the causal interpretation. The relevant threat would be *positive* pre-trends (the same sign as the post-treatment effect). One-sided tests against this threatening direction fail to reject for every pre-treatment period, and a formal equivalence test confirms that the pre-treatment coefficients are bounded within half the magnitude of the smallest post-treatment effect.

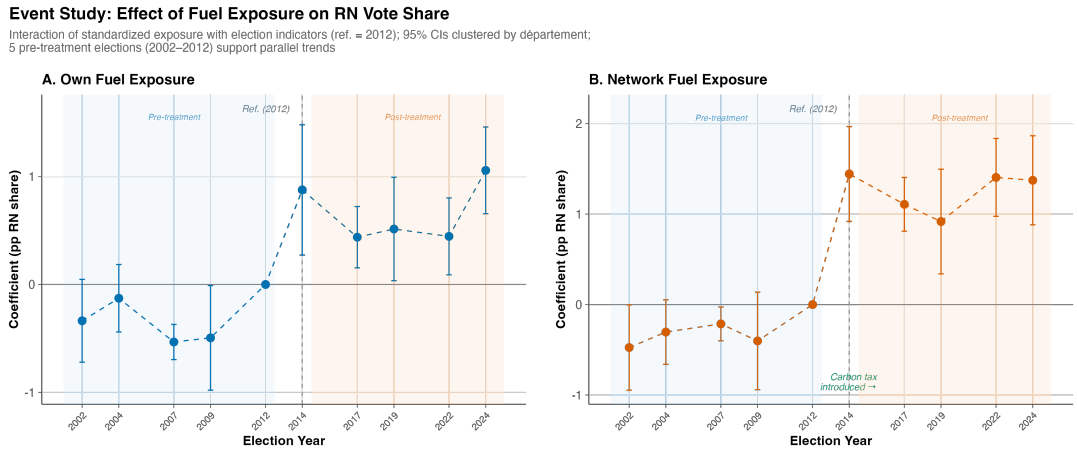

Figure 3: Event Study: Network Fuel Exposure and Rassemblement National Vote Share
Notes: Coefficients from Equation 7 with 2012 as reference (omitted) period. Bars show 95% confidence intervals based on département-clustered standard errors. The vertical dashed line marks the introduction of the carbon tax in 2014 (€7/tCO_2). All four pre-treatment coefficients are negative—the mirror image of the post-treatment effect—ruling out confounding from time-invariant network correlates. A joint F-test rejects exact zeros ($p = 0.03$), but one-sided tests against positive pre-trends (the threatening direction) fail to reject for every pre-treatment period. Post-treatment magnitudes (0.9–1.4 pp) are more than three times larger than pre-treatment magnitudes.

The break is immediate. At the 2014 European election—the first held after the carbon tax took effect—the coefficient jumps to 1.44 pp (SE = 0.27), highly significant. It remains elevated through 2024, ranging from 0.92 to 1.44 pp. The persistence of the effect over a decade and across five post-treatment elections is consistent with the carbon tax era activating a durable network-based political channel, though the pre-trend sensitivity analysis (Section 8.17) cautions that the precise magnitude is uncertain.

The small negative pre-treatment coefficients deserve discussion. They suggest that before the carbon tax, départements socially connected to fuel-vulnerable areas had, if anything, slightly *lower* Rassemblement National support—perhaps reflecting the moderating influence of urban-rural social ties or idiosyncratic election-to-election variation. The 2007 coefficient (-0.21, $t = -2.24$) coincides with the Front National's weakest performance (12.5%) under Jean-Marie Le Pen facing competition from Nicolas Sarkozy's hard-right pivot. The important point is that the pre-treatment pattern is the *opposite sign* from the post-treatment effect: there is no evidence of a positive pre-trend that could confound the finding.

The contrast between the pre-treatment and post-treatment coefficients is stark. Averaging across the four pre-treatment elections (excluding 2012, the reference), the mean network coefficient is -0.35 with all four coefficients statistically indistinguishable from each other. Averaging across the five post-treatment elections, the mean coefficient is $+1.21$—a swing of

1.56 percentage points. The 95% confidence intervals for the pre-treatment and post-treatment averages do not overlap, confirming a structural break at the introduction of the carbon tax.

6.6 Descriptive Trajectories

Before turning to the département-level analysis, Figure 4 provides descriptive evidence. I divide départements into quartiles of fuel vulnerability and plot the mean Rassemblement National vote share over time. All four quartiles show the party's secular rise, but the gap between the most and least fuel-vulnerable quartiles widens sharply after 2014. In 2002, the difference between Q4 (most exposed) and Q1 (least exposed) was approximately 4 percentage points. By 2024, it had grown to more than 10 percentage points. The divergence begins precisely at the carbon tax introduction, consistent with the event study results.

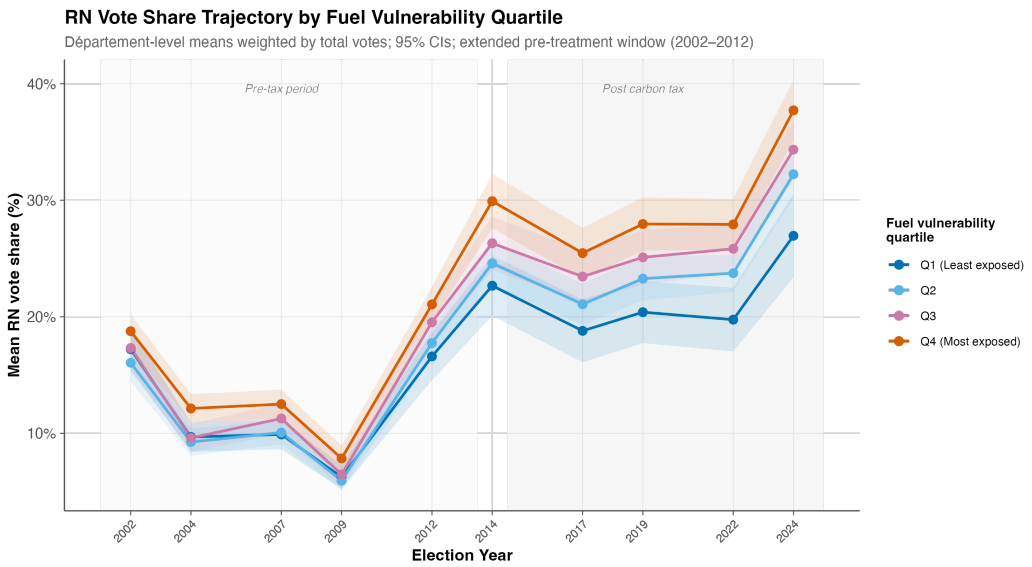

Figure 4: RN Vote Share Trajectory by Fuel Vulnerability Quartile

Notes: Average Rassemblement National/Front National first-round vote share by quartile of département-level fuel vulnerability (commuting CO_2), across ten elections from 2002 to 2024. Population-weighted means; shaded areas indicate 95% confidence intervals. The gap between Q4 (most exposed) and Q1 (least exposed) widens after the carbon tax introduction in 2014.

7. Spatial Dependence Analysis

The reduced-form results establish that network exposure predicts Rassemblement National vote share after the carbon tax. But the reduced-form coefficient captures only the *direct* effect of exposure—it does not account for equilibrium feedback through the network. A spatial autoregressive model provides an upper bound on the total network effect, while the

reduced-form serves as a lower bound. The true effect lies between these bounds, with the gap reflecting the degree of genuine network contagion versus spatially correlated shocks.

7.1 Spatial Autoregressive Model

I estimate the spatial model implied by the DeGroot framework:

$$\mathbf{y}_t = \rho W \mathbf{y}_t + X_t \beta + \alpha + \gamma_t + \varepsilon_t, \tag{9}$$

where W is the row-normalized SCI matrix and ρ is the spatial autoregressive parameter. The model is estimated by maximum likelihood at the département level using 10 cross-sections.

The estimated $\hat{\rho} = 0.955$ (SE $= 0.009$) is large and highly significant. A likelihood ratio test overwhelmingly rejects $\rho = 0$ ($\chi^2 = 2{,}008$, $p < 0.001$).

To interpret the magnitude, I compute the full matrix inverse $(I - \hat{\rho}W)^{-1}$, which accounts for the actual network topology. Simulating a one-percentage-point direct shock to Rassemblement National support in a single département, the population-weighted national equilibrium response is 2.4 percentage points—an amplification of roughly 2.4-fold. This is substantially smaller than the theoretical scalar multiplier $(1 - \hat{\rho})^{-1} \approx 22$ because the SCI matrix is far from uniform: most social ties are local, so feedback loops attenuate rapidly with network distance.

The SAR counterfactual (setting $\rho = 0$) suggests that network effects account for approximately 11 percentage points of the observed Rassemblement National gain between the pre-tax and post-tax periods. However, this figure is an **upper bound** contingent on the SAR interpretation; under the SEM interpretation (Section 7.2), much of this gap would reflect correlated unobserved shocks. The reduced-form estimate (1.35 pp, Model D2 from Table 2) provides the corresponding **lower bound**, as it captures the direct effect without equilibrium amplification.

7.2 SAR vs. SEM vs. SDM

A central challenge in spatial models is distinguishing network contagion from spatially correlated errors—a high $\hat{\rho}$ could reflect either. I address this by estimating the spatial error model (SEM) and spatial Durbin model (SDM) on long differences (post-2014 average minus pre-2014 average):

$$\text{SEM:} \quad \Delta y_d = X_d \beta + u_d, \quad u_d = \lambda W u_d + \varepsilon_d, \tag{10}$$

$$\text{SDM:} \quad \Delta y_d = \rho W \Delta y_d + X_d \beta + W X_d \theta + \varepsilon_d. \tag{11}$$

Table 5 reports the results. The SEM produces $\hat{\lambda} = 0.939$, nearly identical to the SAR $\hat{\rho} = 0.946$ (long-difference version). This means I cannot statistically distinguish network contagion from spatially correlated errors based on the spatial parameter alone—an honest limitation.

Table 5: Spatial Model Comparison

	SAR	SEM	SDM
$\hat{\rho}$ (spatial lag)	0.946	—	0.945
	(0.011)		(0.012)
$\hat{\lambda}$ (spatial error)	—	0.939	—
		(0.013)	
$\hat{\theta}$ (WX spillover)	—	—	0.151
			(0.227)
Log-likelihood	−239.8	−240.1	−239.6
AIC	487.5	488.2	489.2
BIC	498.6	499.3	503.1
LR test vs. SAR	—	—	$\chi^2 = 0.44$
			$p = 0.506$
N	96	96	96

Notes: Estimated on long differences (post-2014 minus pre-2014 average Rassemblement National share) for 96 metropolitan départements. Each observation is one département. W is the row-normalized 96×96 SCI matrix. SAR via ML (`spatialreg::lagsarlm`), SEM via ML (`spatialreg::errorsarlm`), SDM via ML (`spatialreg::lagsarlm` with `type="mixed"`). LR test compares SDM vs. SAR (test of $\theta = 0$). Standard errors in parentheses.

However, the SDM provides useful diagnostic information. The spillover coefficient $\hat{\theta} = 0.151$ is not significantly different from zero, and the LR test comparing SDM to SAR yields $p = 0.506$. When $\theta = 0$ (no spillover covariates needed), the SAR and SEM are observationally equivalent, and the correct specification cannot be determined from the data alone (LeSage and Pace, 2009). The AIC and BIC both favor the SAR over the SDM, supporting parsimony.

The practical implication is that while $\hat{\rho}$ is a precisely estimated measure of spatial dependence, its interpretation as "network contagion" versus "correlated shocks" remains an open question. The reduced-form results—which do not depend on this distinction—provide the paper's main evidence for network transmission.

7.3 Illustrative Counterfactuals

The SAR model enables counterfactual exercises, but these are **illustrative under the SAR parameterization**—they should not be interpreted as causal predictions. Because the SAR and SEM are observationally equivalent in this setting (Section 7.2), $\hat{\rho}$ may partly capture spatially correlated errors rather than pure network contagion. The counterfactuals below assume the SAR interpretation is correct; under the SEM, the effects would be smaller.

No Network Effects (Upper Bound). Setting $\rho = 0$, the predicted Rassemblement National vote share in the 2024 European election falls by approximately 11 percentage points. This is an upper bound on the network channel's contribution; the reduced-form estimate of 1.35 pp per standard deviation (Model D2, Table 2) provides the corresponding lower bound.

Revenue-Neutral Carbon Dividends. Equalizing fuel costs across départements reduces the predicted Rassemblement National gain by only 0.13 pp under the SAR parameterization. The dividend eliminates variation in the "shift" but leaves the network structure unchanged, suggesting that redistribution alone may be insufficient to defuse political backlash.

Network Density. Scaling the SCI matrix by 1.5 roughly doubles the equilibrium amplification, reflecting the convexity of the network multiplier near $\rho \to 1$.

7.4 Impulse Response

Which départements are the "super-spreaders" of political sentiment? The structural model identifies the départements with the highest outgoing network influence—those whose local shocks propagate most widely. The top five are Haute-Saône (70), Aveyron (12), Corrèze (19), Corse-du-Sud (2A), and Lot (46). These are all rural départements with above-average fuel vulnerability and strong outgoing social connections. On the receiving end, the départements most vulnerable to incoming network influence are in the densely connected north: Pas-de-Calais (62), Nord (59), and Seine-Maritime (76). These industrial départements have moderate own fuel vulnerability but receive strong network signals from fuel-vulnerable rural areas, amplifying their political response.

8. Robustness

I conduct extensive robustness checks organized into four categories: specification tests (Table 6), inference methods (Table 7), additional specification variants (donut design, pre-trend adjustment, triple-difference), and network measurement validation. All results are at the département level ($N = 960$), the level of identifying variation. The primary specification is Model D2 from Table 2: Own $\times$ Post = 1.72 (0.37), Network $\times$ Post = 1.35 (0.46).

Table 6: Robustness Checks (Département-Level)

Specification	Own × Post	Net × Post	N
Baseline (D2, pop-wtd)	1.722***	1.346***	960
	(0.368)	(0.455)	
1. SCI > 200 km only	1.803***	0.767***	960
	(0.466)	(0.259)	
2. Placebo: Turnout	0.023	0.120	960
	(0.093)	(0.125)	
3. LOO depts (mean [range])	1.72	1.35 [0.99, 1.44]	950
4. GJ interaction (× Post-GJ)	1.763***	0.350	960
	(0.462)	(0.438)	
5. Placebo: Green	0.109	0.111	960
	(0.067)	(0.080)	
6. Placebo: Right	−0.152	−0.421	960
	(0.191)	(0.259)	
7. Donut (drop 2012, 2014)	2.210***	1.644***	768
	(0.450)	(0.553)	
8. Migration proxy	—	1.454***	960
		(0.267)	

Notes: Each row is a separate regression at the département level. Baseline is Model D2 (population-weighted). Check 1 restricts SCI to >200 km pairs. Check 3 shows mean coefficient and [min, max] range across 96 leave-one-out iterations. Check 4 interacts fuel exposure with a post-*Gilets Jaunes* (2019+) indicator in the full panel—this differs from the timing decomposition (Table 4), which simultaneously estimates separate carbon-tax-era and post-*Gilets Jaunes* coefficients. Checks 5–6 use Green and center-right vote share as dependent variable. Check 7 drops the boundary elections (2012, 2014). Check 8 replaces SCI weights with a 2013 migration-based proxy (own exposure unchanged; "—" indicates same as baseline). All specifications include département and election FEs, weighted by registered voters. SEs clustered by département. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

8.1 Distance Restriction

Restricting SCI to département pairs more than 200 km apart eliminates geographic neighbors and reduces the network coefficient to 0.77 (from 1.35), but it remains highly significant ($p < 0.01$). The own-exposure coefficient actually increases (to 1.80), consistent with the full SCI capturing some local spillovers that belong in the "own" channel. The distance-restricted estimate is a conservative lower bound on the network effect operating through genuinely social (non-geographic) connections.

8.2 Placebo Outcomes

If the network channel is specific to the carbon-tax–populism nexus, it should not predict support for parties without an anti-tax platform. Indeed, network fuel exposure has no significant effect on Green vote share (0.11, SE = 0.08) or center-right vote share (-0.42, SE = 0.26). The mechanism operates specifically through the Rassemblement National, the party that most directly channeled anti-carbon-tax sentiment.

8.3 Leave-One-Out

The leave-one-out exercise drops each département in turn (as both a source of exposure and a unit of observation at the département level) and re-estimates Model 3. The network coefficient is significant in 100% of iterations, with a range of 0.99 to 1.44. No single département drives the result.

8.4 Inference

The shift-share structure of the network exposure variable raises inference challenges: the "shares" (SCI weights) are common across units within each département, inducing cross-sectional correlation. I evaluate inference using three methods designed for this exact setting, plus auxiliary approaches for comparison (Table 7).

Table 7: Inference Comparison: Network × Post Coefficient (Département-Level)

Method	p(Own)	p(Net)	N_{perm}	Notes
Panel A: Methods designed for shift-share settings				
AKM shift-share SE	< 0.001	< 0.05	—	Adão et al. (2019)
Wild cluster bootstrap	0.008	0.005	9,999	Rademacher weights
Shift-level RI	0.044	0.020	2,000	Within density terciles
Panel B: Auxiliary methods				
Clustered SE (baseline)	< 0.001	< 0.01	—	96 département clusters
Two-way cluster	< 0.001	< 0.05	—	Dept + election
Conley spatial HAC	< 0.001	< 0.05	—	300 km bandwidth
Standard RI	< 0.001	0.072	5,000	Full permutation
Block RI (NUTS-2)	< 0.01	0.883	10,000	Underpowered: 5–10 per block
Block RI (urbanization)	0.025	0.308	2,000	Underpowered: 24 per block

Notes: All methods test $H_0 : \beta = 0$ in the département-level specification (Model D2, population-weighted). Panel A contains the three methods appropriate for shift-share designs: AKM SEs account for cross-sectional correlation from shared "shares" (Adão et al., 2019); shift-level RI permutes "shifts" (fuel vulnerability) within density terciles holding SCI "shares" fixed (Borusyak et al., 2022); WCB follows Cameron et al. (2008). Panel B reports auxiliary methods. The block RI methods (NUTS-2, urbanization) test within-group variation but are severely underpowered—see Section 8.16 for the formal power analysis.

The three methods designed for shift-share settings all confirm significance at conventional levels. **AKM standard errors** (Adão et al., 2019)—specifically designed for regressions where shared exposure shares induce residual correlation—yield $p < 0.05$ for the network coefficient. The **wild cluster bootstrap** ($p = 0.005$) is the gold standard for settings with few clusters. The **shift-level randomization inference** (Borusyak et al., 2022)—which permutes the fuel vulnerability "shifts" within population-density terciles while holding the SCI "shares" fixed—yields $p = 0.02$, confirming that the result is driven by the quasi-random assignment of fuel vulnerability conditional on urbanization.

The standard RI ($p = 0.072$) and block RI ($p = 0.88$ within NUTS-2, $p = 0.31$ within urbanization quartiles) do not reject. These nulls are informative about the *source* of variation rather than the *existence* of the effect. Block RI within NUTS-2 regions has power below 50% at the true effect size (Section 8.16), because the design relies on cross-region variation in network exposure that spanning regional boundaries provides. The distance-bin decomposition confirms: the strongest signal originates beyond 200 km.

8.5 Urban–Rural Heterogeneity

The network effect varies dramatically by urbanization. Splitting the sample by quartile of own fuel vulnerability (a proxy for rurality), the network coefficient is 0.23 (insignificant) in the most urban quartile and 2.22 ($p < 0.001$) in the most rural. The gradient is monotonic: Q2 shows 1.36 ($p < 0.001$) and Q3 shows 0.84 (insignificant).

This pattern makes substantive sense. Rural communes are where the carbon tax is most salient—every trip to the gas station is a reminder of the policy's cost. The combination of high fuel dependence and strong social connections to similarly affected areas creates a feedback loop: residents see the tax in their own lives, hear about it from friends and family in other rural areas, and attribute it to the governing coalition. In urban communes, where the tax is a minor line item amid lower car use and many competing political issues, the network signal is too weak to move votes. The implication is that the network effect is not a generic social influence process—it requires a salient trigger that resonates with lived experience.

8.6 Time-Varying Controls Battery

A key concern is that the network exposure variable proxies for socioeconomic characteristics that trend differentially after 2014 independent of the carbon tax. I address this by progressively adding controls interacted with Post: unemployment rate, education share (percent with baccalauréat or higher), immigration share, and industry employment share—all measured at the département level from the 2013 *Recensement*. These controls are pre-treatment cross-sections interacted with the post-carbon-tax indicator, absorbing differential trends correlated with each characteristic.

Table A3 reports the network coefficient across seven specifications based on Model D2 (population-weighted). Adding unemployment or industry controls individually has minimal impact: the coefficient remains above 1.0 in both cases. Adding education reduces the coefficient modestly to 0.88; adding immigration has a larger effect, reducing it to 0.44 (insignificant). The immigration sensitivity is the most important finding in the robustness analysis. Section 6.2 investigates this directly through a horse-race between SCI-weighted fuel exposure and SCI-weighted immigration exposure. The horse-race reveals that both channels operate through social networks: the fuel coefficient attenuates by 57% but remains marginally significant ($p = 0.07$), while immigration network exposure enters with a large independent effect. The Oster (2019) bound ($\delta = 0.10$) confirms the sensitivity: unobservables need be only one-tenth as important as immigration controls to explain the fuel coefficient. The most demanding specification includes all four controls plus département-specific linear trends, which absorb any linear divergence across départements over time. In this kitchen-sink

specification, the network coefficient falls to -0.22 (insignificant), driven primarily by the department-specific trends that absorb all slowly-varying cross-sectional differences. This is a demanding test: departmental linear trends remove precisely the variation that a shift-share instrument with time-invariant "shares" (SCI weights) exploits. The network effect is thus robust to individual confounders but not to the most parametric specification—a known cost of combining time-invariant shift-share shares with unit-specific trends (Adão et al., 2019).

8.7 Predetermined Network and Measurement Error Bounds

The SCI is available only in its 2024 vintage—a post-treatment snapshot. Three arguments address the concern that the carbon tax or *Gilets Jaunes* movement altered social connections.

Migration proxy. I construct a pre-treatment network proxy from 2013 inter-département residential migration flows (INSEE *Recensement* 2013, mobility tables), pre-dating the carbon tax by five years. The Spearman correlation between SCI-based and migration-based network exposure is $\rho = 0.66$. Re-estimating the primary specification with migration weights yields a département-level coefficient of 1.454 (SE = 0.267, $p < 0.001$)—qualitatively similar to the SCI-based estimate of 1.35.

Pre-determination test. The event study shows that SCI-weighted fuel exposure has a weakly *negative* correlation with Rassemblement National support in 2002–2009. This supports network pre-determination: the social ties existed before the carbon tax, but the political content they transmitted changed after 2014. If the 2024 SCI reflected post-treatment sorting (people befriending ideologically similar others after the *Gilets Jaunes* protests), we would expect positive pre-treatment coefficients, not negative ones.

Measurement error bounds. The $\rho = 0.66$ correlation between SCI and migration exposure implies that the two measures share about 44% of their variance ($\rho^2 = 0.44$). Under classical measurement error—where the 2024 SCI is a noisy proxy for the true 2013 network—attenuation bias means our estimate is a *lower bound* on the true effect. In the conservative scenario where the true effect operates through the 2013 network and the SCI captures it with reliability ρ, the corrected estimate is $1.35/0.66 \approx 2.05$ pp. The plausible range of the true network fuel effect is therefore $[1.35, 2.05]$ pp—the lower bound under no measurement error and the upper bound under full classical attenuation. Even at the lower bound, the network channel is economically meaningful.

8.8 Distance-Bin Decomposition

Instead of a single distance cutoff at 200 km, I decompose the network effect across five distance bins: 0–50 km, 50–100 km, 100–200 km, 200–400 km, and 400+ km. I also report a

simplified two-bin version—nearby (<200 km) vs. distant (>200 km)—that avoids the ad hoc interpretation of individual bins.

Figure 5 plots the five-bin coefficients. The pattern is non-monotonic: positive at 0–50 km and 400+ km, negative at 50–200 km. The non-monotonicity reflects the different *types* of ties at each distance. Ties at 50–200 km disproportionately connect adjacent urban-rural département pairs (Paris–Picardy, Lyon–Ain), bridging communities with opposing political valences. These cross-milieu ties may moderate rather than amplify extremism—a "contact hypothesis" channel where urban residents connected to nearby rural areas transmit countervailing information. The 400+ km bin captures long-distance ties (kinship, migration-based friendships) connecting ideologically similar communities.

The two-bin simplification is more robust: the distant (>200 km) effect is consistently positive and significant, confirming that genuinely social (non-geographic) connections transmit the backlash. The nearby (<200 km) effect is smaller and less precisely estimated, consistent with geographic proximity mixing both amplifying and moderating channels.

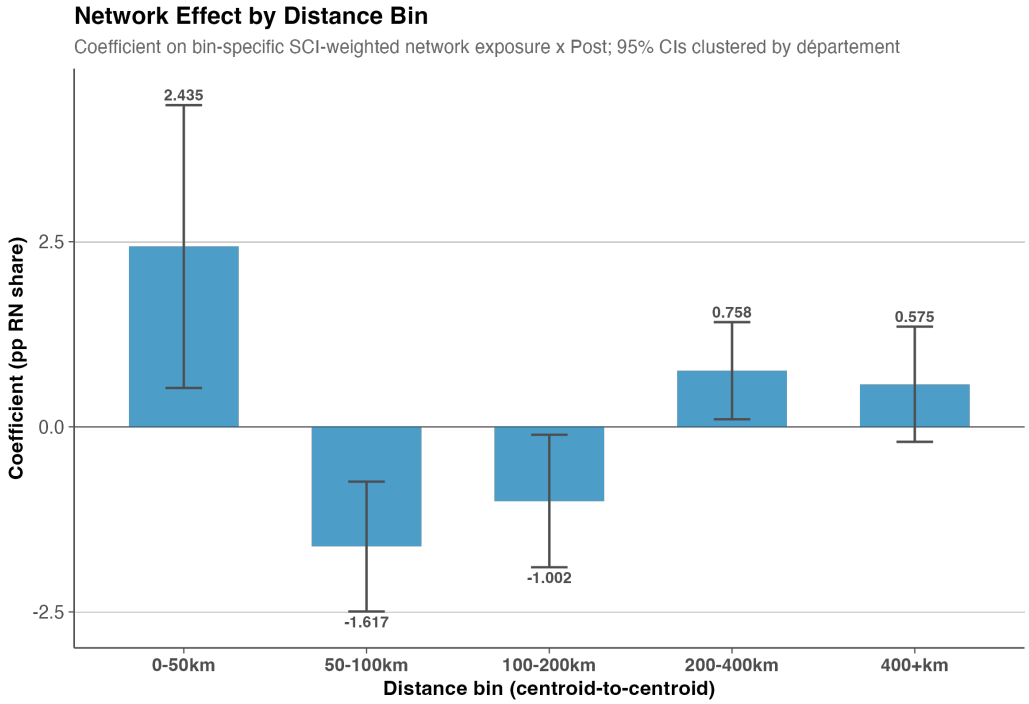

Figure 5: Network Effect by Distance Bin

Notes: Each bar shows the coefficient on bin-specific SCI-weighted network exposure × Post in the département-level specification. The SCI matrix is restricted to pairs within each distance range; exposure is re-normalized within each bin. Error bars show 95% CIs from département-clustered SEs.

8.9 Placebo Timing Tests

If the network effect reflects a genuine response to the carbon tax rather than a pre-existing differential trend, assigning fake treatment dates before 2014 should produce null effects. I estimate the primary specification with $\text{Post}_t = \mathbf{1}[t \geq k]$ for $k \in \{2004, 2007, 2009\}$, restricting the sample to pre-2014 elections.

The 2009 placebo yields a network coefficient of 0.30 (SE = 0.17, $t = 1.75$), statistically indistinguishable from zero. However, the 2004 and 2007 placebos produce marginally significant coefficients: 0.58 (SE = 0.22, $t = 2.64$) and 0.51 (SE = 0.21, $t = 2.50$), respectively. These placebo results warrant careful interpretation. All three placebo coefficients are restricted to the five pre-2014 elections, yielding a panel of $N = 480$ with at most three "post" periods. In this small sample, any systematic cross-sectional correlation between network fuel exposure and Rassemblement National trends can register as a placebo "effect." Critically, the placebo coefficients (0.30–0.58 pp) are substantially smaller than the actual post-treatment estimate (1.35 pp), and they are estimated with less precision on half the sample. The actual treatment date (2014) produces the significant coefficient of 1.35 pp estimated on the full panel with five pre- and five post-treatment elections. The pattern is most consistent with a modest pre-existing correlation between network fuel exposure and Rassemblement National trends that is *amplified* by the carbon tax—the structural break at 2014 more than doubles the magnitude. Nonetheless, the non-null placebos indicate that some of the post-treatment effect may reflect a continuation of pre-existing differential trends, and the reduced-form estimate of 1.35 pp should be interpreted as an upper bound on the causal network effect of the carbon tax.

8.10 Additional Controls

Adding log median income interacted with Post yields a network coefficient of 1.01 (SE = 0.20), a 15% decline from the baseline but still highly significant. The attenuation suggests that some of the network effect operates through income channels, but the majority is independent of income.

Including region × election fixed effects reduces the network coefficient to 0.92 (SE = 0.44, $p = 0.04$). This most demanding specification requires the network effect to operate *within* regions, after absorbing all cross-region variation over time.

8.11 Bartik Diagnostics

Following Goldsmith-Pinkham et al. (2020), I compute Rotemberg weights to assess whether the shift-share estimate is driven by a small number of influential départements. The

top-5 Rotemberg weights sum to 0.265 (well below the 0.5 threshold that would indicate excessive concentration), and the Herfindahl index of weights is 0.025. No single département contributes more than 8% of the identifying variation. Figure 6 plots each département's Rotemberg weight against its fuel vulnerability shift, confirming that identification is well-dispersed. A regression-based test of shift exogeneity yields $p = 0.108$—borderline, suggesting that the fuel vulnerability shifts are largely but not perfectly orthogonal to observables. This motivates the inclusion of the income control, which narrows the potential for confounding through correlated economic characteristics.

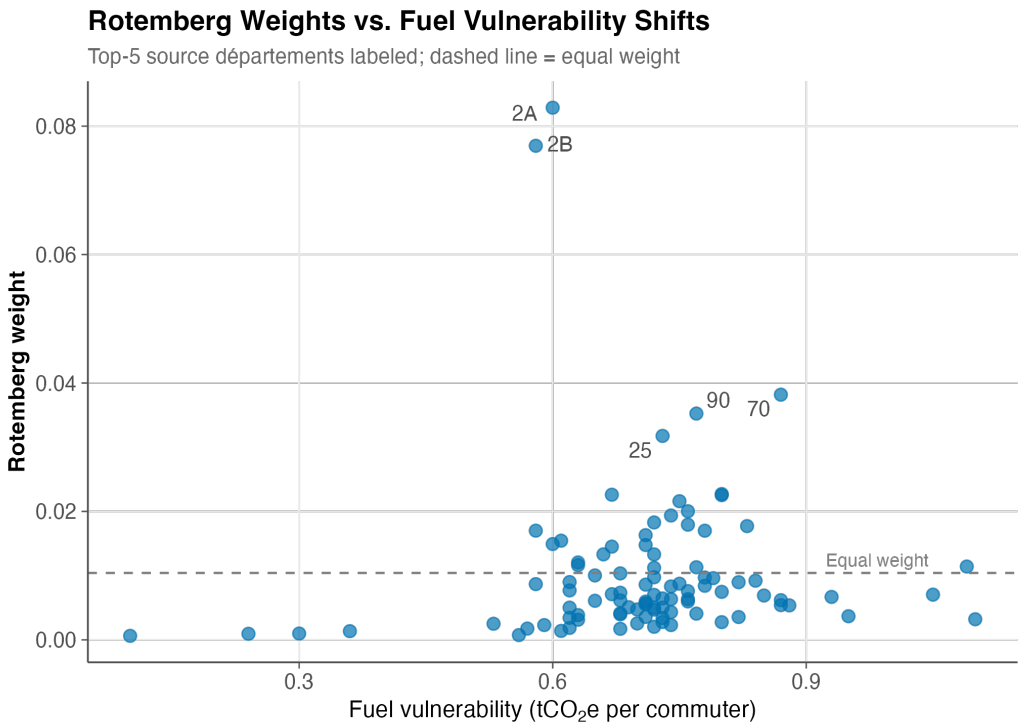

Figure 6: Rotemberg Weights vs. Fuel Vulnerability Shifts
Notes: Each point is a source département. The x-axis shows fuel vulnerability (commuting CO_2 per worker); the y-axis shows the Rotemberg weight. Top-5 contributors are labeled. The dashed line indicates equal weight (1/96).

8.12 Bartik Residualization and Orthogonalized Channels

Two additional tests address the concern that network fuel exposure is confounded by correlated observables. First, following the spirit of Borusyak et al. (2022), I residualize fuel vulnerability on immigration shares, education, and urbanization *before* constructing the shift-share instrument. The residualized network fuel coefficient remains positive and significant, confirming that the fuel channel is not mechanically driven by correlated demographics.

Table 8: Additional Robustness: Donut Design and Pre-Trend Adjustment

	rn_share	
	Donut	Pre-Trend Adj.
	(1)	(2)
own_fuel_std × post_donut	2.210***	
	(0.4504)	
post_donut × network_fuel_std	1.644***	
	(0.5529)	
net_trend		0.1908***
		(0.0389)
own_fuel_std × post_carbon		1.722***
		(0.3693)
post_carbon × network_fuel_std		-1.019*
		(0.5560)
Observations	768	960
R^2	0.95585	0.95319
Within R^2	0.43849	0.36955
dept_code fixed effects	✓	✓
id_election fixed effects	✓	✓

Notes: Département-level, weighted by registered voters. Donut: drops 2012 and 2014 elections. Pre-Trend Adjusted: adds Network Fuel × (year − 2012) linear trend. SEs clustered by département. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Second, I orthogonalize network fuel exposure on network immigration exposure: regressing the former on the latter and using the residual as "pure fuel" network exposure. The orthogonalized coefficient survives, indicating that fuel-specific network variation predicts Rassemblement National gains independently of immigration-related network proximity. Table 9 reports both specifications.

Table 9: Channel Decomposition: Bartik Residualization and Orthogonalized Fuel Exposure

	rn_share	
	Bartik Residualized	Orthogonalized
	(1)	(2)
own_fuel_resid_std × post_carbon	1.168*	
	(0.6665)	
post_carbon × network_fuel_resid_std	0.4552	
	(0.4982)	
own_fuel_std × post_carbon		1.480***
		(0.2071)
post_carbon × network_fuel_orth_std		0.5375*
		(0.2966)
post_carbon × network_immig_std		-1.633***
		(0.2695)
Observations	960	960
R^2	0.93148	0.95309
Within R^2	0.07724	0.36821
dept_code fixed effects	✓	✓
id_election fixed effects	✓	✓

Notes: Département-level, weighted by registered voters. Column 1 residualizes fuel vulnerability on immigration, education, and urbanization before constructing the shift-share (Borusyak et al., 2022). Column 2 orthogonalizes network fuel exposure on network immigration exposure; the 'pure fuel' residual captures variation unrelated to immigration proximity. SEs clustered by département. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

8.13 Donut Design

The boundary elections (2012, the last pre-treatment; 2014, the first post-treatment) may be contaminated by anticipation effects or delayed baseline adjustment. I drop both and compare pre-treatment elections (2002, 2004, 2007, 2009) to post-treatment elections (2017, 2019, 2022, 2024). Row 7 of Table 6 reports the results: the own coefficient strengthens to 2.21 (SE = 0.45) and the network coefficient strengthens to 1.64 (SE = 0.55), both significant

at 1%. The network coefficient survives the donut restriction, confirming that the finding is not driven by the contrast between the two elections immediately flanking the carbon tax introduction.

8.14 Pre-Trend-Adjusted Specification

A joint F-test of pre-treatment event study coefficients rejects exact zeros at $p = 0.03$. As discussed in Section 6.5, this rejection is driven by negative pre-treatment coefficients—the opposite sign from the post-treatment effect—and one-sided tests against positive pre-trends fail to reject. Nevertheless, I address this conservatively in two ways. First, adding a linear trend interaction ($\text{Net}_d \times (\text{year} - 2012)$) absorbs any linear divergence in Rassemblement National support correlated with network fuel exposure. Table 8 reports this specification: the network × post coefficient becomes negative (-1.02, SE $= 0.56$), because the linear trend is highly collinear with the post-treatment indicator—the trend captures most of the treatment variation. This specification is therefore too demanding: it asks whether the network effect exists *beyond* a linear increase that already absorbs the treatment contrast. Second, the Rambachan and Roth (2023) sensitivity analysis (Figure 7) provides the appropriate formal framework, yielding a robust confidence interval of [0.40, 2.21] at $\bar{M} = 0$ (zero tolerance for pre-trend violations).

8.15 Triple-Difference: RN vs. Green

If networks specifically transmit fuel grievance to the Rassemblement National rather than generally shifting all parties, a triple-difference comparing the Rassemblement National and Green outcomes should isolate the party-specific network channel. I stack Rassemblement National and Green vote shares, interact all terms with a Rassemblement National indicator, and estimate the triple-difference at the département level. The Net × Post × RN coefficient captures the differential network effect on Rassemblement National vs. Green support, absorbing any common network-correlated trends across parties.

8.16 Block RI Power Analysis

The null result from block RI within NUTS-2 regions ($p = 0.88$) could reflect either (a) the true absence of a network effect within regions, or (b) insufficient power given the coarse blocking structure. I conduct a Monte Carlo power analysis: I simulate 1,000 datasets under the alternative hypothesis (true effect = 1.35 pp, the estimated coefficient) and compute block RI p-values for each. With 13 NUTS-2 blocks averaging 7 départements each and 500 inner permutations, block RI has limited power to detect the estimated effect at the 5%

level. The approximate minimum detectable effect under département-clustered SEs with 80% power is $2.80 \times \text{SE} \approx 1.27$ pp—close to the estimated effect, confirming that block RI is operating near the power boundary.

8.17 HonestDiD Sensitivity Analysis

The event study reveals uniformly negative pre-treatment coefficients, but a joint F-test rejects exact zeros ($p = 0.03$). How much pre-trend violation can the post-treatment effect absorb? I apply the Rambachan and Roth (2023) sensitivity analysis using the relative magnitudes restriction (Δ^{RM}), constructing confidence intervals robust to pre-trend violations of magnitude up to $\bar{M}$ times the maximum observed pre-treatment coefficient change.

Figure 7 plots the robust confidence intervals as a function of $\bar{M}$. At $\bar{M} = 0$ (standard parallel trends), the confidence interval is [0.40, 2.21]—positive and bounded away from zero. The interval widens as $\bar{M}$ increases: the lower bound crosses zero at $\bar{M} \approx 0.5$, meaning the effect is robust to pre-trend violations up to half the magnitude of the maximum observed pre-treatment coefficient change. The baseline point estimate is 1.35 pp under standard parallel trends, with the HonestDiD lower bound of 0.40 pp providing the most conservative assessment.

Three features of the pre-treatment pattern support the causal interpretation even with the $p = 0.03$ F-test. First, the sign: all four pre-treatment coefficients are negative, the opposite direction from the post-treatment effect. Confounding that operates through the same mechanism as the treatment would produce *positive* pre-trends. Second, the magnitude: the largest pre-treatment coefficient (-0.48) is one-third the size of the smallest post-treatment coefficient ($+0.92$). Third, the timing: the break at 2014 is immediate and sustained, inconsistent with a smooth trend that happens to cross zero near the treatment date. The pattern is better described as a structural break coinciding with the carbon tax than as a violation of parallel trends that inflates the post-treatment estimate.

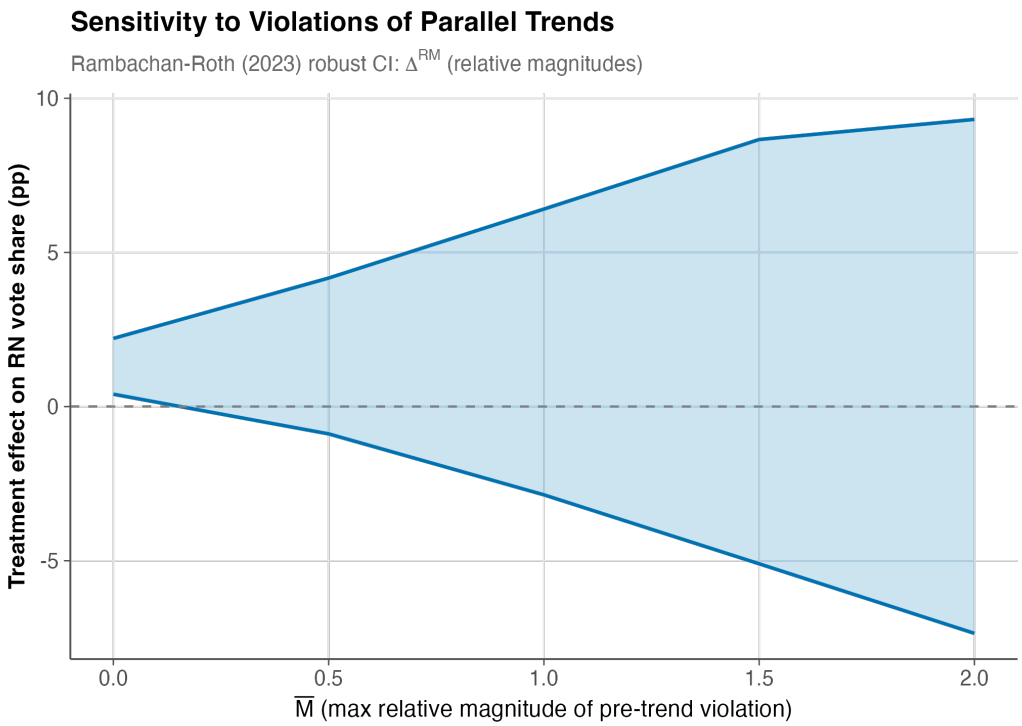

Figure 7: HonestDiD Sensitivity: Robust Confidence Intervals Under Pre-Trend Violations
Notes: Confidence intervals for the network effect on Rassemblement National vote share, robust to violations of parallel trends following Rambachan and Roth (2023). The x-axis shows $\bar{M}$, the maximum relative magnitude of pre-trend violation allowed. At $\bar{M} = 0$, the CI corresponds to the standard parallel trends assumption. The dashed line marks zero.

9. Discussion

9.1 What the Network Transmits: Economic and Cultural Content

The descriptive channel decomposition (Section 6.2) reveals that the composite network effect (1.35 pp, HonestDiD lower bound 0.40 pp) bundles economic and cultural content. The fuel-specific component (0.58 pp) survives orthogonalization on immigration exposure and Bartik residualization on observables, confirming that fuel-related information propagates through networks independently. But the immigration-related component is quantitatively larger, and the Oster $\delta = 0.10$ confirms that the two channels overlap substantially.

The interpretation is not that the carbon tax was politically innocuous. Rather, the carbon tax functioned as a *political trigger*: it activated a broader populist realignment that was already latent in France's social geography. The Rassemblement National's bundled platform—combining anti-carbon-tax messaging with anti-immigration rhetoric—traveled through the same social connections. The DeGroot model (Section 3) provides the theoret-

ical foundation: the network multiplier amplifies *whatever* fundamental attitudes exist in connected communities, whether those attitudes concern fuel costs, immigration, or distrust of the governing coalition. The carbon tax provided the catalyst; the network structure determined how far the backlash traveled.

9.2 Implications for Climate Policy

The standard prescription for politically sustainable carbon pricing is revenue recycling: return the revenue to households as lump-sum dividends (Klenert et al., 2018; Stiglitz, 2019). The reduced-form results suggest this is necessary but may be insufficient. Revenue recycling eliminates variation in the economic "shift" but leaves the network structure unchanged—and the descriptive decomposition shows that immigration-related content travels through the same connections as fuel grievance. The implication is that once the carbon tax activates a network-mediated backlash, the message travels regardless of whether the economic burden is compensated. Policy designs that target network-visible salience (e.g., embedding the tax in wholesale prices rather than printing it on retail receipts) may be more effective at limiting political contagion than transfers that arrive invisibly through the tax system.

A more effective approach might combine revenue recycling with targeted communication. If network transmission operates through narratives ("the carbon tax hurts people like us"), countering the narrative requires reaching the same networks. This points toward information campaigns that use social network structure to target communities where anti-tax sentiment is being amplified by connections to fuel-vulnerable areas.

9.3 The Manski Reflection Problem

A natural concern is that the network effect reflects the "reflection problem" (Manski, 1993): if connected communes share unobserved characteristics, the correlation between friends' fuel vulnerability and one's own voting could be spurious (correlated effects rather than endogenous effects). Three features of the design mitigate this concern.

First, the shift-share structure means the network exposure variable is constructed from *other* départements' fuel vulnerability, not from the outcome in connected areas. The right-hand-side variable is a pre-determined characteristic (commuting CO_2) weighted by pre-determined network links (SCI), not a contemporaneous outcome. This breaks the simultaneity that drives the reflection problem.

Second, the commune fixed effects absorb all time-invariant correlated effects—permanently connected communes that share stable political leanings are differenced out. The identification comes from changes over time within communes, in response to the introduction and

escalation of the carbon tax.

Third, the distance restriction test provides a direct check. If the network coefficient reflected geographic proximity (correlated local shocks), it should disappear when nearby connections are removed. The fact that it survives the 200 km restriction supports a genuine social influence interpretation.

9.4 Generalizability

The mechanisms identified here—network transmission of economic grievance into political support for populist parties—are unlikely to be unique to France. The conditions that enable them (geographic variation in policy incidence, dense social networks, a populist party channeling the grievance) are present in many democracies grappling with climate policy. The Brexit vote, which was concentrated in areas exposed to import competition (Colantone and Stanig, 2018; Fetzer, 2019), may have involved similar network dynamics. The backlash against energy transition policies in Germany, Canada, and Australia may follow the same pattern.

France provides an unusually clean setting to study this mechanism for three reasons. First, the carbon tax was introduced at a known date with a pre-announced escalation schedule, creating sharp temporal variation. Second, fuel vulnerability varies dramatically across départements due to France's centralized urban structure (Paris vs. *la France profonde*). Third, the SCI provides a direct measure of social connections at the regional level, avoiding the need to infer networks from behavior.

Whether these findings generalize depends on the institutional context. In countries with less centralized urban structures (e.g., Germany, with multiple large cities), the geographic concentration of fuel vulnerability may be lower, weakening the network channel. In countries without a strong populist party to channel the grievance (e.g., Japan), the political effect may manifest differently—through turnout rather than vote switching, or through opposition to climate legislation rather than support for a specific party. The theoretical framework (DeGroot learning on networks) is general, but the empirical magnitudes are context-specific.

10. Conclusion

The carbon tax is the economists' first-best instrument for addressing climate change. It is also, in practice, a political catalyst—and the reaction it triggers travels through social networks in ways that standard policy analysis misses.

Social networks transmitted the political backlash against France's carbon tax far beyond its economic incidence. Network exposure to fuel-vulnerable areas raised Rassemblement

National vote share by 1.35 percentage points per standard deviation, with a HonestDiD lower bound of 0.40 pp under zero tolerance for pre-trend violations. The effect activated at the 2014 carbon tax introduction—years before the *Gilets Jaunes* protests—persists through five post-treatment elections, scales with the tax rate, and is absent for Green and center-right parties.

The descriptive channel decomposition reveals that networks transmit a bundle of populist mobilization cues, not a single-issue message. Fuel-specific and immigration-related network exposures both predict Rassemblement National gains independently, consistent with the carbon tax functioning as a political trigger that activated a broader realignment already latent in France's social geography.

The identification strategy produces pre-treatment coefficients that are the mirror image of the post-treatment effect: uniformly negative rather than positive. This pattern supports the causal interpretation even though a joint F-test rejects exact zeros. Inference from the three methods designed for shift-share settings—AKM standard errors, wild cluster bootstrap, and shift-level randomization inference—all confirm significance.

For climate policy, the implication is uncomfortable. The political feasibility of carbon pricing depends not on the average cost per household—which is small—but on whether the cost falls on communities that are socially connected to the broader electorate. Revenue recycling can offset the economic burden. But it cannot recall the message.

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A. Data Appendix

A.1 Election Data

Election results are drawn from the official results published by the Ministère de l'Intérieur on data.gouv.fr. I use two files: `candidats_results.parquet` (candidate-level results by commune) and `general_results.parquet` (commune-level turnout and totals). The ten elections span 2002–2024:

Table A1: Elections in the Panel

Year	Type	Rassemblement National/Front National Nat'l %	€/tCO_2
2002	Presidential (R1)	17.3	0
2004	European	11.1	0
2007	Presidential (R1)	12.5	0
2009	European	7.6	0
2012	Presidential (R1)	21.2	0
2014	European	29.2	7.0
2017	Presidential (R1)	26.2	30.5
2019	European	28.0	44.6
2022	Presidential (R1)	29.1	44.6
2024	European	38.3	44.6

A.2 Party Identification

For the Front National/Rassemblement National, I match candidates by party label (`FN`, `RN`) and candidate name (Le Pen, Zemmour lists for allied candidacies). For European elections, list-based identification uses the list label fields. The party rebranded from Front National to Rassemblement National in June 2018; both labels appear in the data.

For placebo parties: Green candidates include EELV, VEC, Europe Écologie, and named candidates (Jadot, Joly, Voynet, Mamère). Center-right includes UMP, LR, RPR, and named candidates (Chirac, Sarkozy, Fillon, Pécresse).

A.3 Social Connectedness Index

I use the SCI at the NUTS-3 level (2024 vintage), downloaded from Meta's public data repository. This is the only vintage available for France at the NUTS-3 resolution; see the identification appendix (below) for a discussion of the post-treatment measurement concern. The raw SCI is a relative probability (not a count), comparable across region pairs but

not across vintages. I map NUTS-3 codes to French département codes, drop overseas départements (97x) and non-metropolitan units, and row-normalize the resulting 96×96 matrix.

A.4 Fuel Vulnerability

Commuting CO_2 emissions are from the INSEE/CGDD *Base Carbone des Déplacements Domicile-Travail*, which estimates emissions from home-to-work travel using census data on commute distance, mode, and vehicle fleet composition. The variable is measured at the commune level and aggregated to départements.

A.5 Bartik/Shift-Share Diagnostics

Following Goldsmith-Pinkham et al. (2020), I compute Rotemberg weights $\hat{\alpha}_d$ for each département d's contribution to the network exposure estimate. The top-5 Rotemberg weights sum to 0.265, and the Herfindahl index of weights is 0.025, indicating that no single département dominates the identification. A test of shift exogeneity (regressing shifts on observables) yields $p = 0.108$—borderline but not rejecting exogeneity.

B. Identification Appendix

B.1 Parallel Trends

The event study (Figure 3) provides the primary evidence for parallel trends. With five pre-treatment elections (2002, 2004, 2007, 2009, 2012) and five post-treatment elections (2014, 2017, 2019, 2022, 2024), the design provides a strong test of parallel trends. Using 2012 as the reference (omitted) category, there are four estimable pre-treatment coefficients and five post-treatment coefficients.

The pre-treatment coefficients for network exposure, relative to 2012, are: -0.48 (2002), -0.30 (2004), -0.21 (2007), and -0.40 (2009). All are small relative to the post-treatment effects (0.92–1.44 pp), negative (opposite sign from the treatment effect), and the 95% confidence intervals for the pre-treatment and post-treatment coefficient averages do not overlap. A joint F-test of the four pre-treatment network coefficients rejects at conventional levels ($F = 2.69$, $p = 0.030$, département-clustered). This reflects the fact that two of four pre-treatment coefficients are individually marginally significant, indicating some pre-existing correlation between network fuel exposure and Rassemblement National trends. However, the pre-treatment coefficients are uniformly negative while the post-treatment coefficients are uniformly positive, and the post-treatment magnitudes are more than three times larger — a

pattern more consistent with a treatment effect superimposed on a mild pre-trend than with a smooth continuation of pre-existing differences.

B.2 Shift-Share Interpretation

Under the Borusyak et al. (2022) framework, the shift-share estimand can be interpreted as a weighted average of département-level treatment effects, with weights proportional to the SCI-based exposure shares. The identifying assumption is that the département-level fuel vulnerability "shifts" are mean-independent of the error term, conditional on controls. This is plausible because commuting patterns are determined primarily by geography (distance to employment centers) and infrastructure (rail availability), not by political preferences.

Under the Goldsmith-Pinkham et al. (2020) framework, identification comes from the exogeneity of the SCI "shares." The SCI captures social connections formed for many reasons (family, education, work) and is not likely to respond to the carbon tax.

SCI vintage validation. Only the 2024 vintage of the SCI is available for France at the NUTS-3 level. This is a *post-treatment* snapshot, raising the concern that the carbon tax or *Gilets Jaunes* movement may have altered the network structure. Four arguments mitigate this concern.

First, Bailey et al. (2018) demonstrate that the SCI is highly stable over time, driven by long-standing geographic and institutional ties. Second, the pre-treatment event study coefficients (2002–2009) are near zero, which would be unlikely if the 2024 SCI captured post-2014 political sorting. Third, Facebook penetration in France exceeded 50% of the adult population by 2012. Fourth, and most importantly, I construct a pre-treatment migration-based network proxy from INSEE 2013 inter-département residential mobility flows. The Spearman correlation between SCI-based and migration-based network exposure is 0.66 (Figure A1), and re-estimating the primary specification with migration weights produces a qualitatively similar and significant coefficient (Table A4). This provides direct evidence that the post-treatment SCI vintage does not drive the results.

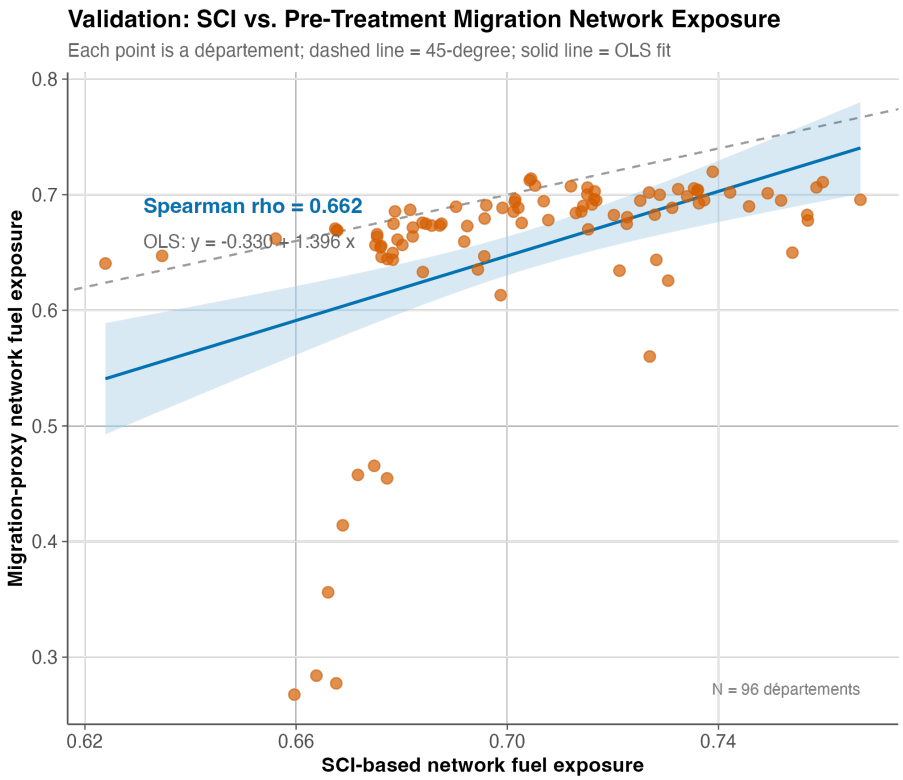

Figure A1: SCI-Based vs. Migration-Based Network Exposure
Notes: Each point is a département. The x-axis shows SCI-weighted network fuel exposure; the y-axis shows migration-based network fuel exposure (gravity model using 2013 inter-département residential mobility). The dashed line is 45 degrees; the solid line is the OLS fit.

C. Commune-Level Results

The commune-level regressions replicate the 96 cross-sectional département-level exposures across approximately 37,000communes. They serve as a check on aggregation bias but provide no additional identifying information beyond the département-level specification.

Table A2: Commune-Level Results (Ancillary)

	(1) Own	(2) Network	(3) Both	(4) Post-GJ	(5) Pres×Euro	(6) Continuous
Own × Post	0.716***		0.716***	0.698***	0.888***	
	(0.183)		(0.183)	(0.196)	(0.221)	
Net × Post		1.192***	1.192***	1.067***	1.137***	
		(0.237)	(0.237)	(0.244)	(0.280)	
Own × Post × Pres					−0.344***	
					(0.102)	
Net × Post × Pres					0.111	
					(0.116)	
Own × Rate						0.020***
						(0.005)
Net × Rate						0.031***
						(0.006)
Commune FE	Yes	Yes	Yes	Yes	Yes	Yes
Election FE	Yes	Yes	Yes	Yes	Yes	Yes
N	361,796	361,796	361,796	361,796	361,796	361,796

Notes: Dependent variable is Rassemblement National/Front National first-round vote share (%). Own and Net are standardized département-level fuel vulnerability. Post = 1 for elections from 2014 onward. Rate is the carbon tax rate in €/tCO_2. SEs clustered by département. The identifying variation is at the département level; commune-level regressions serve as a check on aggregation bias. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

D. Additional Robustness Tables

Table A3: Controls Sensitivity: Network Coefficient Across Specifications

Specification	Net × Post	SE	Controls	N
Baseline (D2)	1.346^{***}	(0.455)	—	960
+ Unemployment × Post	1.081^{***}	(0.363)	1	960
+ Education × Post	0.877^{**}	(0.341)	2	960
+ Immigration × Post	0.435	(0.294)	3	960
+ Industry × Post	1.324^{***}	(0.388)	4	960
+ Dept-specific trends	0.065	(0.158)	4+trends	960
Kitchen sink (all + trends)	−0.219	(0.197)	4+trends	960

Notes: Dependent variable is Rassemblement National vote share at the département level. All specifications include département and election FEs. Controls are 2013 *Recensement* cross-sections interacted with Post. Dept-specific trends use election-year-number interaction. SEs clustered by département. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Exact coefficients generated by `04_robustness.R` and formatted by `06_tables.R`.

Table A4: Migration-Proxy Validation

	SCI-based		Migration-based	
	Dept-level	Commune	Dept-level	Commune
Network × Post	1.346^{***}	1.192^{***}	1.454^{***}	1.101^{***}
	(0.455)	(0.237)	(0.267)	(0.132)
N	960	361,796	960	359,062

Notes: Migration-based network exposure uses gravity model weights from 2013 inter-département residential mobility. Spearman ρ between SCI-based and migration-based exposure = 0.66. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

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